OR-AS Operations Research Applications and Solutions	Case Name: Apartment Building Finishing Works (1)	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs
		Risk Analysis	Random simulation One of nine std. scenarios User defined distributions
Submitted by	William & Jef		
Date	2016	Project Control	Automatic tracking Tracking based on user input
File Name	C2016-31 Apartment Building Finishing Works (1)		

1. Project description

Project authenticity

The project is a construction plan for building the structural part of an apartment building, mainly focusing on concrete elements like columns and walls. It runs from March to July 2016, lasts about 85 working days, and has a total cost of around €1.32 million. The work is organized into tasks with set durations, costs, and dependencies, and the project also includes basic tracking tools to monitor progress, costs, and performance during construction.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	23
Planned Duration (PD)	105
Budget At Completion (BAC)	488,936.00€
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	31%
Activity Distribution (AD)	40%
Length of Arcs (LA)	0%
Topological Float (TF)	11%

* standard eight-hour working days=

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0.00	0.00	N/A
CRI-rho	100.00	0.00	N/A
CRI-tau	100.00	0.00	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	-	-	-
CRI-rho	-	-	-
CRI-tau	-	-	-

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	35	46	0.68
SI	54	35	0.42
SSI	13	20	1.29
CRI-r	17	14	1.11
CRI-rho	17	14	1.09
CRI-tau	11	9	1.11

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	3.09	1.41
PV - SPI	4.37	2.63
PV - SCI	4.37	2.63
ED - 1	3.05	1.38
ED - SPI	4.30	2.56
ED - SCI	4.30	2.56
ES - 1	4.11	-2.09
ES - SPI(t)	8.53	-4.24
ES - SCI(t)	8.53	-4.24

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.00	0.00
CPI	0.00	0.00
SPI	1.43	1.21
SPI(t)	3.94	-1.65
SCI	1.43	1.21
SCI(t)	3.94	-1.65
0.8 CPI + 0.2 SPI	0.37	0.33
0.8 CPI + 0.2 SPI(t)	0.64	0.00

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-6689.25	-39711.25	-6.75	0.96	0.83	0.88	1
std dev	1901.08	45244.10	3.78	0.03	0.19	0.08	0
final	-9537.00	0.00	-12	0.98	1.00	0.90	1

2.3.3.2. Time forecasting

PD	105

Real Duration	117

Late	11.43%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	157.04	14.06	34.22	-34.22
PV - SPI	184.45	50.08	57.65	-57.65
PV - SCI	192.94	60.39	64.90	-64.90
ED - 1	159.77	9.41	36.56	-36.56
ED - SPI	188.95	45.91	61.49	-61.49
ED - SCI	194.78	56.21	66.48	-66.48
ES - 1	154.63	5.68	32.16	-32.16
ES - SPI(t)	167.00	16.23	42.74	-42.74
ES - SCI(t)	171.78	23.08	46.82	-46.82

2.3.3.3. Cost forecasting

BAC	488,936.00€

Real Cost	498, 473. 00 €

Over budget	1.95%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	495625.25	1901.08	0.57	0.57
CPI	507742.05	18245.70	2.10	-1.86
SPI	585270.71	152339.74	17.41	-17.41
SPI(t)	536083.77	60475.57	7.55	-7.55
SCI	605059.12	185844.38	21.38	-21.38
SCI(t)	551458.53	85384.37	10.63	-10.63
0.8 CPI + 0.2 SPI	518541.35	36082.50	4.03	-4.03
0.8 CPI + 0.2 SPI(t)	512744.27	25473.45	2.98	-2.86